MILITARY SPECIFICATION

PROCESSOR, SIGNAL, FIRE AND FLIGHT AIR DATA SUBSYSTEM, HELICOPTER ARMAMENT: M143

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification contains performance and environmental requirements, quality assurance provisions, and packaging for the Processor, Signal, Fire and Flight Air Data Subsystem, Helicopter Armament: M143.

2. APPLICABLE DOCUMENTS.

2.1 Government documents. Unless otherwise specified, the following Specifications and Standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-E-6051	Electromagnetic Compatibility Requirements, Systems
MIL-I-8500	Interchangeability and Replaceability of Components Parts for Aircraft and Missiles

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-F-13926	Fire Control Materials: General Specification Governing the Manufacture and Inspection of
MIL-M-38510	Microcircuits, General Specification for
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication and Radar for Aerospace and Control Systems, Procedure for
MIL-S-46844	Solder Bath, Soldering and Printed Wiring Assemblies, Automatic Machine Type

STANDARDS

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-810C	Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT, RESEARCH AND DEVELOPMENT CENTER (ARDC)

11776549	Processor, Signal, Fire and Flight Air Data Subsystem: M143
11776552	Installation Drawing, Processor Signal, Fire and Flight Air Data Subsystem, Helicopter Armament: M143

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict of this specification and the references cited herein, the text of this specification shall take precedence.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D3951 Standard Practices for Commercial Packaging

3. REQUIREMENTS.

3.1 Materials. Materials shall be in accordance with the applicable drawings, specifications, and standards.

3.2 Parts. All parts shall comply with applicable drawings, specifications, and standards.

3.3 Air Data System Electronic Processor (EPU). The EPU shall comply with all the requirements specified on drawing 11776549, all associated drawings, and with all the requirements specified in applicable specification and standards. The EPU covered by this specification shall be configured for use on AH-1S series aircraft. The EPU shall process signals from the AH-1S Helicopter. The processed signals shall be sent to the Low Airspeed Indicator (LAI), drawing 11776301, the Fire Control Computer: M22, and the AN/ASN-128 Doppler Navigation System.

3.3.1 Major components list. Major components of the EPU are:

Nomenclature	Drawing
Pressure Transducer Unit	11776576
Analog Interface Unit	11776663
Central Processor Unit	11776653
Power Supply Unit	11776551
Front Panel Assembly	11776673
Chassis	11776666
Rear Cover	11776670

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3.4 Characteristics.

3.4.1 Electrical.

3.4.1.1 Signal interface. The EPU signal interface with the AH-1 series helicopter and with the LAI and AADS shall be as defined in Appendix A to this specification and Figure 1.

3.4.1.1.1 Detail characteristics. The ADS communicates with the Fire Control Computer, Digital: M22 (FCC). Data transmission to the FCC is by means of a dedicated serial-data bus.

3.4.1.1.2 Fire control computer interface. The ADS transmits data to the FCC by means of a dedicated serial data bus. The bus consists of a data line-pair which carries the data and its logical complement, a clock line-pair which carries the associated data clock and its complement, and additional wires which carry the discrete signals required for protocol. The wiring configuration is shown in Figure 2. A timing diagram, illustrating typical signal timing, is shown in Figure 3.

3.4.1.1.3 Bus characteristics. The data line-pair between the ADS and the FCC consists of a twisted shielded wire pair with a characteristic impedance of 71 ohms, ±10 percent. The shield on this pair is grounded in accordance with good EMI suppression practices. Data shall be transmitted and received by differential 3-state logic transmitters and receivers. The signal levels used shall be TTL compatible.

3.4.1.1.4 Data clock characteristics. This signal is a square wave with a 50-percent duty cycle such that the 0-1 transition defines the beginning of a data bit interval. It is sent on a twisted shielded wire pair with the same electrical characteristics as those of the data line. The data clock signal levels shall be TTL compatible. The data clock shall have a nominal pulse rate of 10 Khz.

3.4.1.1.5 Discrete line characteristics. Discrete signals used for protocol shall be transmitted on shielded wire pairs. The signal shall be received differentially. Discrete signals not used for protocol shall be transmitted on single shielded wires.

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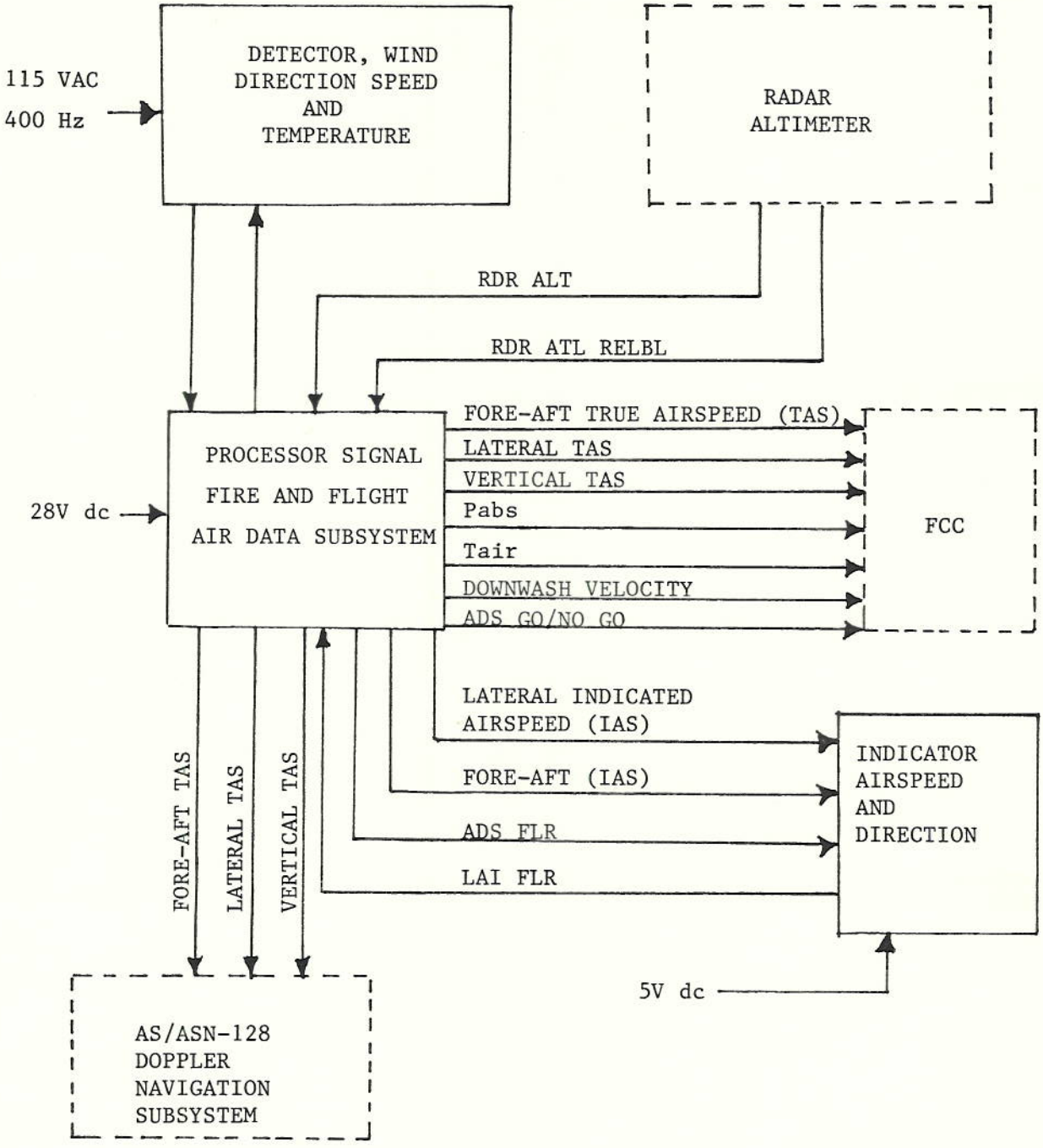


FIGURE 1. EPU interface block diagram

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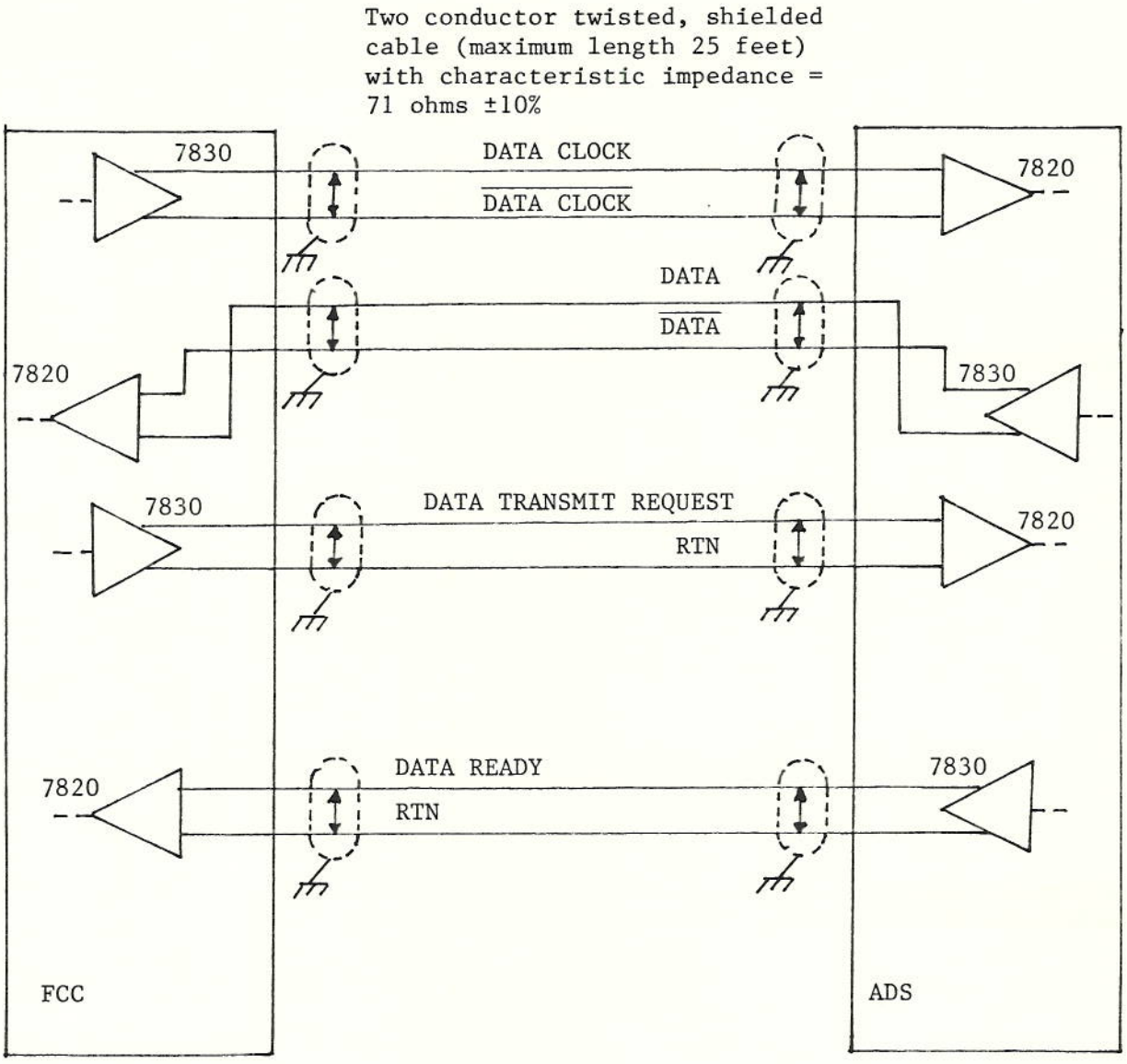


FIGURE 2. Data bus wiring configuration

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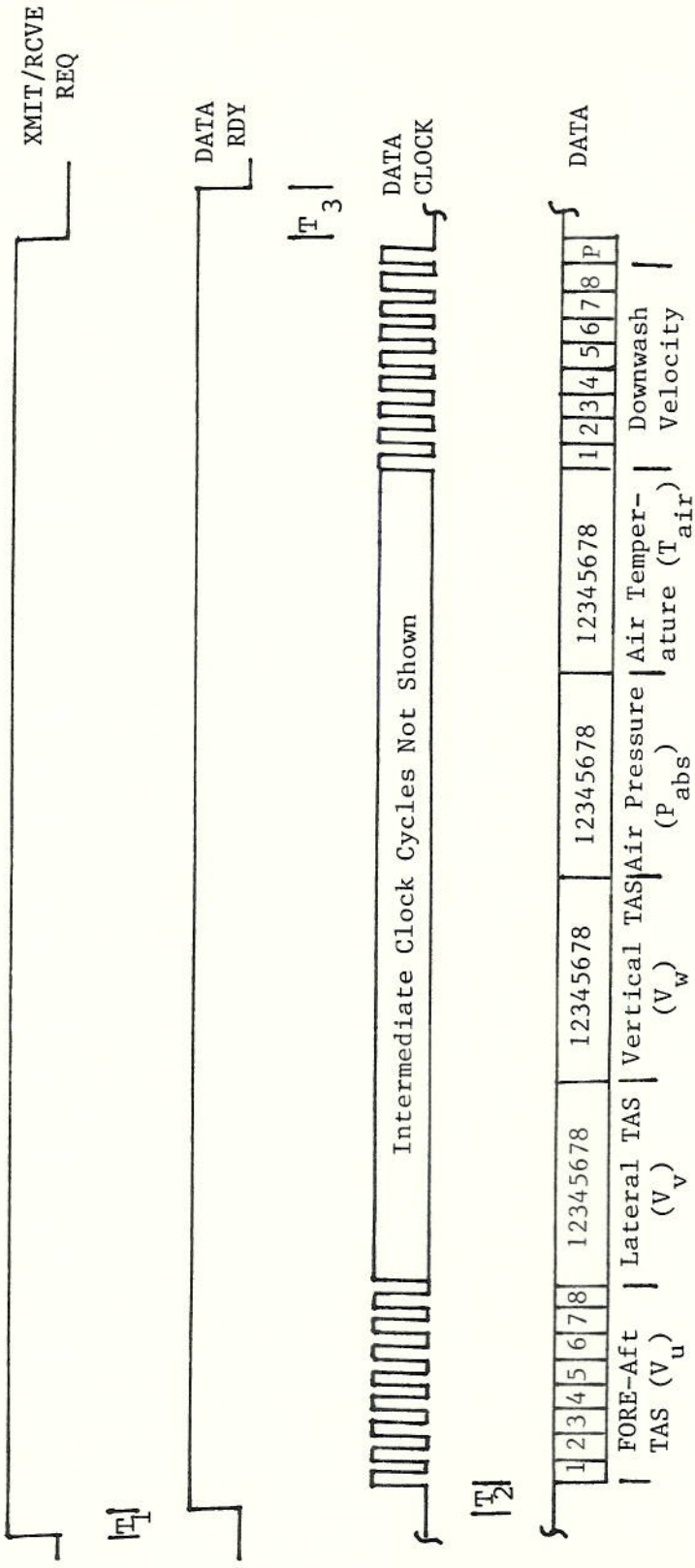


FIGURE 3. Data bus timing diagram

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3.4.1.1.6 Data format. The data shall be sent as serial 8-bit words in binary non-return-to-zero (NRZ) -L code. Each data bit shall begin coincident with the rising edge of the data clock signal supplied by the FCC. Each data word shall form a part of a 49-bit transmission period consisting of six 8-bit words and a trailing parity bit. The least significant bit of each 8-bit word shall be the first bit transmitted for each word. Transmitted parity of the 49-bits shall be odd.

3.4.1.1.7 Signal protocol. Transmission of data along the bus is under control of the FCC. The FCC initiates data transfer from ADS to FCC by supplying a logic 1 to the data transmit request line. After a time, t_1 (minimum = 1 usec, maximum = 250 usec), the ADS responds by supplying a logic 1 to the data ready line. After delay, t_2 (minimum = 1 usec, maximum = 20 msec), the FCC begins sending the data clock signal. The ADS then transmits the data as described in 3.2.1.1.8. At the conclusion of the 49th bit (parity), the FCC returns the data transmit request line to logic 0 and interrupts the data clock. After time delay t_3, not to exceed 1.0 usec, the ADS returns the ready data line to logic 0, and data transfer is complete.

3.4.1.1.8 Data word format. The information sent in the data words shall be in the sequence V_u, V_v, V_w, P_s, T_{air}, and V_{dw}. The ADS shall send all six words each time a data transmit request is received.

3.4.1.2 Electrical power.

3.4.1.2.1 Steady state. The electrical service conditions will be in accordance with MIL-STD-704, Category "B" utilization equipment, except that the transient frequency limit of Figure 5, Limit 4 in MIL-STD-704 shall be 240 Hz. The equipment shall meet all specified requirements of MIL-STD-704 for the types of power specified herein, and shall give specified performance when supplied with electrical power having these characteristics and limits. The power required shall not exceed 28V dc, 34 Watts.

3.4.1.2.2 Electrical power interruptions. Momentary power interruptions to the equipment will be experienced when external primary power is switched to helicopter power or when switching from one helicopter power bus to another. The EPU shall meet the performance requirements of this specification after such momentary power interruptions have been experienced. The EPU shall return automatically to specified performance within 5 seconds after any power interruption.

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3.4.1.2.3 Polarity 'reversal'. Components that receive 28 Volt dc helicopter power shall not be damaged when subjected to a polarity reversal due to inadvertant reversal of the helicopter, 28V dc power. Component operation shall be normal when normal polarity is restored.

3.4.1.2.4 Bonding resistance. The resistance between any of the four bonding points and any exposed metal surfaces of the EPU shall be less than 0.1 ohms.

3.4.1.2.5 Insulation resistance. The insulation resistance shall be greater than 20M ohms at least 10 seconds after 50V dc is applied between all pins of the connector linked together and the EPU bonding point. The insulation resistance shall be greater than 20M ohms at least 10 seconds after 50V dc is applied between pins 17, 25, 26, 34, 35, 43, 51, and 63 connected together and all other pins of the connector linked together.

3.4.2 Performance.

3.4.2.1 Operation. The EPU is used to process data received from the AH-1 Helicopter and the Air Speed and Direction Sensor (AADS). The processed data is used to compute armament aiming and flight information,

3.4.2.2 Input parameter. The EPU shall accept data in accordance with Table I.

3.4.2.3 Output parameter. The EPU shall provide the air data parameters over the ranges and within the error tolerance in Table II.

TABLE I. Input parameters

Parameter	Type	Parameter Range	Voltage Range	Equiv
Sin Alpha	Analog	+1 to -1	9.6V P-P	$\pm 2^{o}$
Cos Alpha	Analog	+1 to -1	9.6V P-P	$\pm 2^{o}$
Sin Alpha-Beta	Analog	+1 to -1	9.6V P-P	$\pm 2^{o}$
Cos Alpha-Beta	Analog	+1 to -1	9.6V P-P	$\pm 2^{o}$
Air Temperature	Resistance	374.4 ohm to 609.9 ohm	N/A	0.6 ohm
Radar Altitude	Analog	0 to 1500 ft	0 to-10.5 Vdc	
Ps	Pnuematic	9.625 to 16 psia	N/A	N/A
Pd	Pnuematic	9.625 to 16 psia	N/A	N/A

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TABLE II. Parameter range of measurement

Parameter	Range	Test Conditions	Required Accuracy
V_u	-32 to +223kn	$-32kn < V < +175kn$	$\pm$4kn
		$-54^oC < T_{AIR} < +71^oC$	
V_v	-32 to +31.75kn	$-54^oC < T_{AIR} < +71^oC$	$\pm$3kn
V_w	-40 to +23.75kn	$-54^oC < T_{AIR} < +71^oC$	$\pm$3kn
P_{ABS}	+9.625 to +16.0 psia	$-25^oC < T_{AIR} < +71^oC$	$\pm$0.06psi
		$-54^oC < T_{AIR} < -25^oC$	$\pm$0.12psi
T_{AIR}	-55° to $+72.5^oC$	$-54^oC < T_{AIR} < +71^oC$	$\pm 3^oC$
V_{DW}	0 to +63.75kn	$-54^oC < T_{AIR} < +71^oC$	$\pm$3kn
V_{IU}	-50 to +50kn	$-54^oC < T_{AIR} < +71^oC$	$\pm$3kn
V_{IV}	-50 to +50kn	$-54^oC < T_{AIR} < +71^oC$	$\pm$3kn

NOTES:

V_u = Fore-aft true airspeed

V_v = Lateral true airspeed

V_w = Vertical true airspeed

P_{ABS} = Absolute (static) air pressure

T_{AIR} = Free-stream temperature

V_{IV} = Lateral indicated airspeed

V_{IU} = Fore-aft indicated airspeed

V_{DW} = Downwash velocity

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3.4.2.4 Fault detection/location. The EPU shall have a Built-In Test (BIT) capability to check the operational status of the Air Data Subsystem (ADS). Bit shall obviate the need for any external test equipment for the performance of pre-flight or in-flight testing. The EPU shall provide detection/identification of a fault to the defective ADS LRU (EPU, LAI, AADS).

3.4.2.5 BITE Output. The BIT circuitry shall also provide a go/no-go output signal to drive a remote indicator. This output shall be as specified in Appendix A.

3.4.2.6 Malfunction indication. A status indicator shall be provided for each LRU. The status indicator shall be a go/no-go device and shall be such as to hold the last result if power is interrupted or removed, and shall be on the surface of the EPU, visible to the maintenance personnel when the equipment is installed in the aircraft.

3.4.2.7 Accuracy. The EPU accuracy values shown in Table II shall be defined as 95 percent confidence values. EPU measured accuracy shall be within the specified accuracies in at least 19 out of each 20 separate measurements to achieve the EPU performance envelope.

3.4.2.8 System turn-on time. The EPU response time from power-on actuation to ready state shall not exceed 5 seconds.

3.4.2.9 Air leakage. Air leakage from the pitot and static lines within the EPU shall not cause an equivalent error greater than 0.5 knot in the dynamic pressure sensed by the AADS. This shall apply for any point within the performance envelope of the EPU, and for the environmental conditions specified herein.

3.4.3 Physical characteristics.

3.4.3.1 Dimensions. Dimensional requirements applicable to envelope size, mounting, and mating shall be in accordance with drawing 11776552.

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3.4.3.2 Weight. EPU weight shall be 7 pounds 8 ounces maximum.

3.4.4 Preconditioning (burn-in). Each EPU procured under this specification shall be subjected to a burn-in test as defined in Figure 4.

3.4.5 Environmental conditions (first article testing only). The EPU shall operate and meet the performance requirements of 3.4.2 under any of the environmental conditions, or a reasonable combination of these conditions, as specified herein.

3.4.5.1 Natural environment. The EPU shall operate and withstand storage in the environments specified herein.

3.4.5.1.1 Altitude. Operating: 0-20,000 feet; nonoperating and storage; 0-50,000 feet.

3.4.5.1.2 Temperature. Operating, nonoperating, and storage: -50° to $+71^{o}C$.

3.4.5.1.3 Humidity. Operation while being subjected to MIL-STD-810, Method 507.1, Procedure II. In Step 6, measurements shall be taken during the last five hours of the last test cycle.

3.4.5.1.4 Sand and dust. Operation after being subjected to MIL-STD-810, Method 510.1, Procedure I, except that the air velocity in Steps 1, 2, and 3 shall be 200 $\pm$ 100 fpm. Step 3 may proceed immediately after stabilization in Step 2, if desired.

3.4.5.1.5 Salt atmosphere. Operation after being subjected to MIL-STD-810, Method 509.1. Operation of the EPU shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surface.

3.4.5.1.6 Fungus. No evidence of viable fungus on any surfaces after being subjected to MIL-STD-810, Method 508.1.

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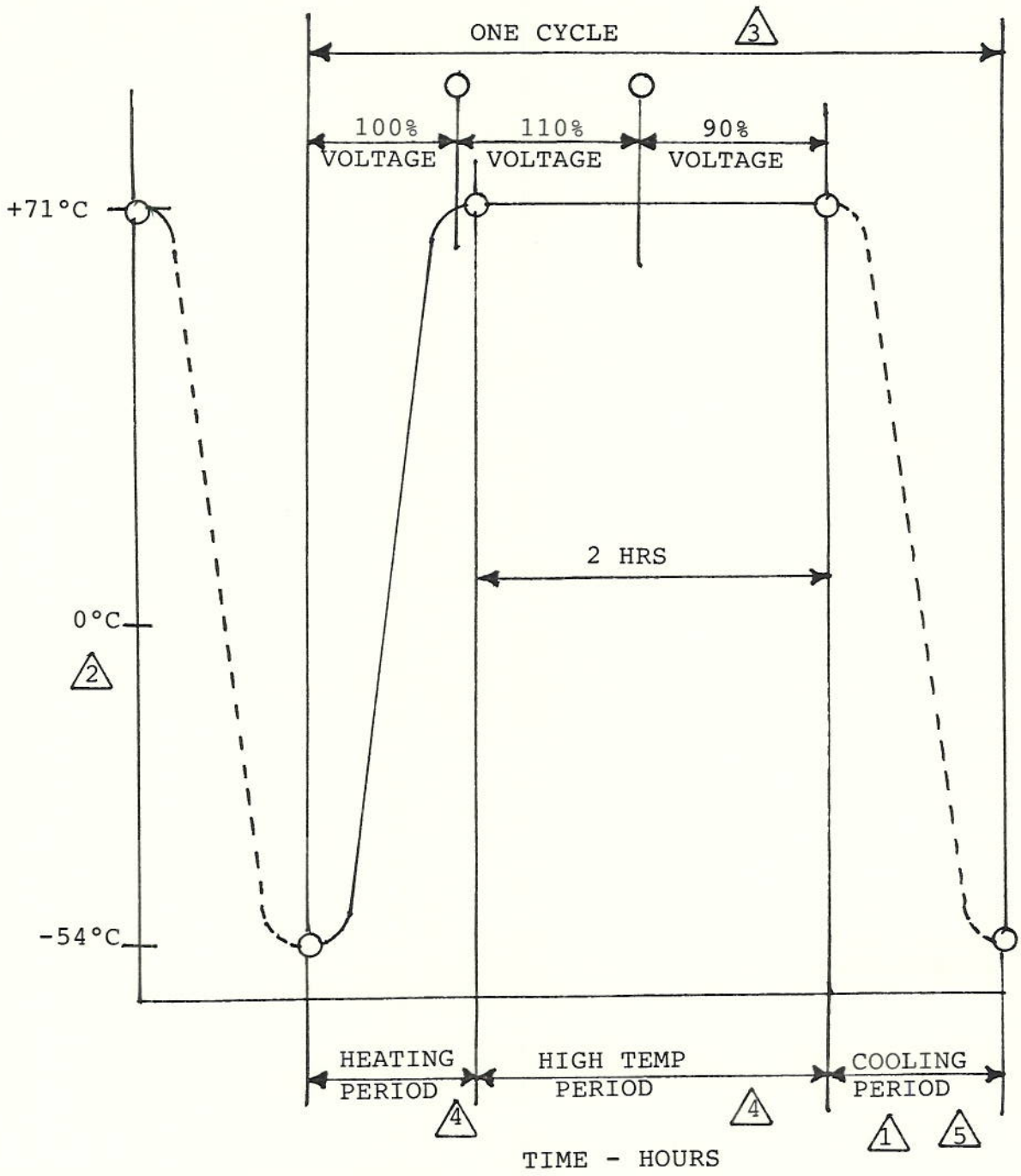


FIGURE 4. Typical temperature cycle - electronic and avionic equipment - Continued

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NOTES:

1. During the cooling period, the chamber temperature shall not be lower than -60°C.

2. Once every 24 hours at 0°C, moisture shall be injected until the equipment being tested is coated with frost.

3. One temperature cycle shall start with the equipment stabilized at the low temperature, heated and stabilized at the high temperature, maintained at the high temperature for 2 hours, then end with the equipment cooled and stabilized at the low temperature. During the temperature cycle, the equipment shall be operated in accordance with its normal duty cycle.

4. Equipment ON; input voltage at 110% and 90%, respectively during each 1/3 of the total time of heating period plus high temperature period; momentarily OFF between input voltage changes.

5. Equipment OFF.

6. Vibration -2.2g ±10% peak acceleration value at any nonresonant frequency between 20 and 60 Hz measured at the mounting points of the equipment. The vibration shall be at least 10 minutes during each hour of operating time.

7. Transition rate between temperature levels shall average 5°C per minute.

FIGURE 4. Typical temperature cycle - electronic and avionic equipment

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3.4.5.1.7 Explosive atmosphere. Operation after being subjected to MIL-STD-810, Method 511.1, Procedure I, except test to be limited to a maximum altitude of 20,000 feet.

3.4.5.1.8 Rain. Operation after being subjected to MIL-STD-810, Method 506.1, Procedure I. During each thirty minute rain period, the wind shall be maintained for fifteen minutes, commencing five minutes after the start of the period as specified in the procedures. The EPU shall be operated during the last ten minutes of the thirty minute period.

3.4.5.2 Service environment. The EPU shall withstand the service environment as specified in MIL-E-5400 for Class 2 equipment, exept as noted in paragraphs 3.4.5.1.1 thru 3.4.5.1.8.

3.4.5.2.1 Bench handling. The EPU shall withstand shock environments encountered in maintenance and servicing.

3.4.5.3 Induced environment.

3.4.5.3.1 Vibration. The EPU shall not sustain damage or subsequently fail to provide specified performance when subjected to the vibration environment defined by Method 514.2, Procedure I, Part I, Table 514.2-III of MIL-STD-810; except that the frequency shall be limited to 500 Hz (reference Figure 5).

3.4.5.3.2 Shock. The EPU shall meet the shock requirements of MIL-STD-810, Method 516.2, Procedure I, Figure 516.2-2 and Method 516.2, Procedure III.

3.4.6 Transportability. The EPU shall withstand vibration environments encountered during transportation as specified in MIL-STD-810, Method 514.2, Procedure X.

3.4.7 Processes.

3.4.7.1 Soldering. All soldering shall be done in accordance with MIL-S-45743 or MIL-S-46844.

3.4.7.2 Microelectronic devices. Microelectronic devices shall conform to MIL-M-38510.

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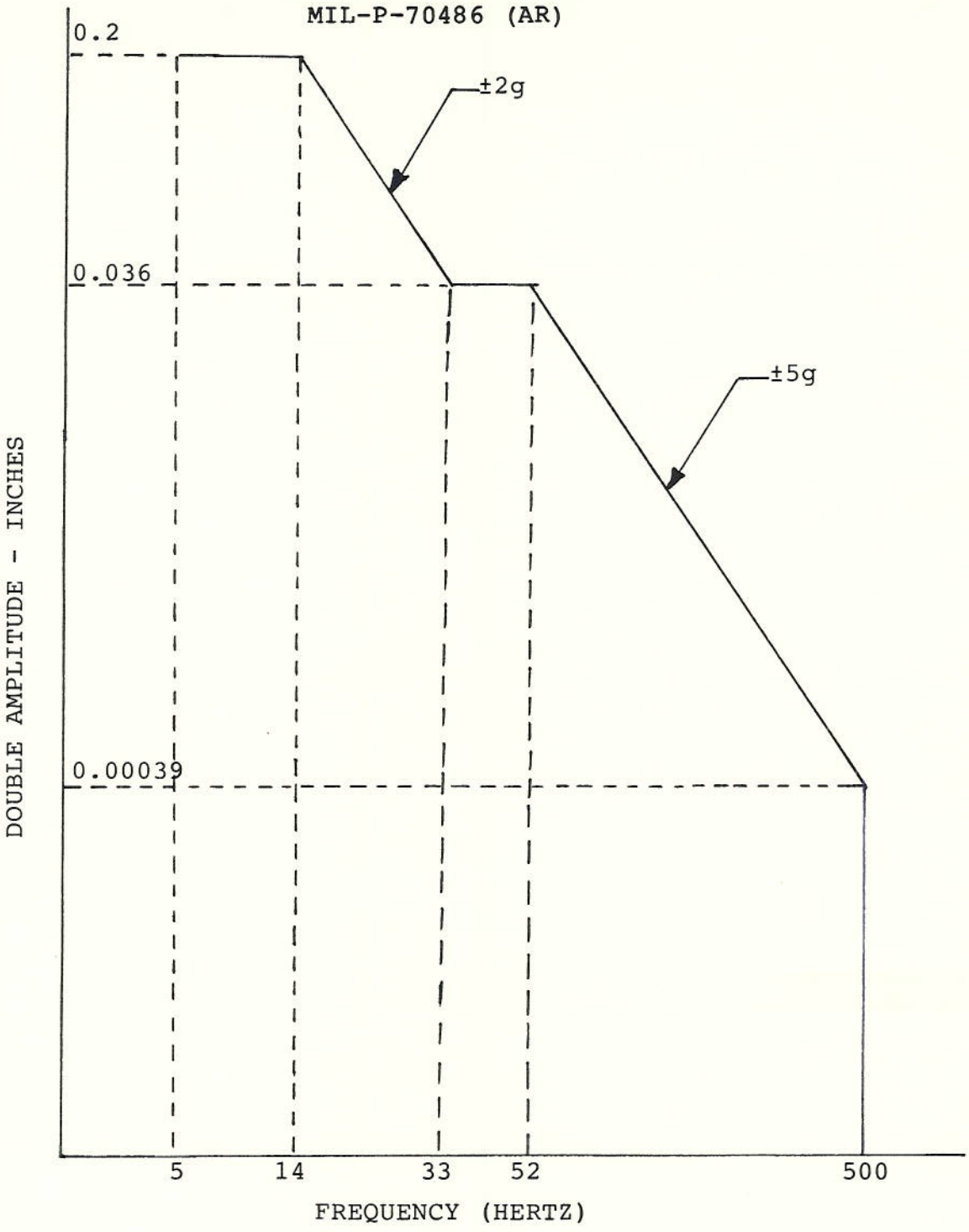


FIGURE 5. Vibration amplitude for qualification test

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3.4.8 Electromagnetic radiation (first article testing only).

3.4.8.1 Electromagnetic compatibility. Each EPU component as installed in the helicopter, shall not cause undesirable response, malfunction, or degraded performance to any other equipment/subsystem installed in or associated with the helicopter; nor shall any component malfunction, exhibit degraded performance or undesirable response beyond the tolerances specified in MIL-E-6051 when other equipment on board the helicopter are singly or collectively operated and performing their intended functions.

3.4.8.2 Electromagnetic interference. The EPU shall meet the requirements of MIL-STD-461 equipment when tested to the following requirements: CE01, CE03, CE04, CE05, CS01, CS02, CS06, RE02, RS01, RS03, RS03.1, and RS02.

3.4.8.3 Circuit isolation. Signal and power potentials shall be electrically isolated from each other and the equipment chassis.

3.4.9 Interchangeability (first article only). All components having the same manufacturer's part number shall be dimensionally and functionally interchangeable in accordance with MIL-I-8500. The item identification and part number requirements of DOD-STD-100 shall govern the manufacturer's part number and changes thereto. Interchangeability must be maintained within the detail parts to the level of replaceable items.

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article by the contractor shall be as specified in the contract.

3.6 Workmanship. Workmanship shall be in accordance with MIL-F-13926 and electronic and mechanical weapon-build practices to ensure safety, proper and dependable operation, and service life. Particular attention shall be given to assure thoroughness of assembly, alignment of parts, electrical connections, cleanliness, tightness of screws and bolts, marking of parts, application of protective finish, and freedom from burrs and rough or sharp edges.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classed as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926.

4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three EPUs selected at random by the Government representative from the first fifteen production EPUs.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in Table III and IV.

4.4.3 Failure. Failure of any EPU to meet the requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any EPU to comply with any stated requirement.

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TABLE III. First article tests

Item	Characteristic	Requirement	Test Procedure
1	Natural environment	3.4.5.1	4.6.4.1
2	Altitude	3.4.5.1.1	4.6.4.2
3	Temperature	3.4.5.1.2	4.6.4.3
4	Humidity	3.4.5.1.3	4.6.4.4
5	Sand and dust	3.4.5.1.4	4.6.4.5
6	Salt atmosphere	3.4.5.1.5	4.6.4.6
7	Fungus	3.4.5.1.6	4.6.4.7
8	Explosive atmosphere	3.4.5.1.7	4.6.4.8
9	Rain	3.4.5.1.8	4.6.4.9
10	Interchangeability	3.4.9	Applicable drawings, visual, MIL-I-8500, & MIL-STD-100
11	Service environment	3.4.5.2	4.6.9.10
12	Bench handling	3.4.5.2.2.1	4.6.4.10.1
13	Induced environment	3.4.5.3	4.6.4.11
14	Vibration	3.4.5.3.1	4.6.4.11.1
15	Shock	3.4.5.3.2	4.6.4.11.2
16	Electromagnetic radiation	3.4.8	4.6.4.12
17	Transportability	3.4.6	4.6.4.11.3

TABLE IV. Performance tests

Item	Characteristic	Requirement	Test Procedure
101	V_u Fore-aft true airspeed	3.4.2.3	4.6.2.11
102	V_v Lateral true airspeed	3.4.2.3	4.6.2.11
103	V_w Vertical true airspeed	3.4.2.3	4.6.2.11
104	P_{ABS} Absolute (static) air pressure	3.4.2.3	4.6.2.10
105	T_{AIR} Free stream temperature	3.4.2.3	4.6.2.7
106	V_{IV} Lateral indicated airspeed	3.4.2.3	4.6.2.9
107	V_{IV} Fore-aft indicated airspeed	3.4.2.3	4.6.2.9
108	V_{DW} Downwash velocity	3.4.2.3	4.6.2.11
109	Fault detection/location	3.4.2.4	4.6.2.4, 4.6.2.5, & 4.6.2.6
110	Air leakage	3.4.2.9	4.6.2.2 & 4.6.2.3

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TABLE V. General tests

Item	Characteristic	Requirement	Test Procedure
301	Fabrication and General Specifications	3.1, 3.2 & 3.3	Applicable Drawings, Visual, and MIL-E-5400, Visual

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

4.5 Quality conformance inspection.

4.5.1 Performance inspection.

4.5.1.1 Sample and tests. Each EPU in every lot shall be subjected to the tests in Table IV and shall be examined visually for completeness, improper assembly, and evidence of poor workmanship.

4.5.1.2 Acceptance. Where any EPU fails to meet any specified requirement, the defective EPU shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.2 General inspection.

4.5.2.1 Sample and tests. One EPU, as a control sample, shall be selected at random by the Government representative from each 100 EPUs produced and shall be subjected to the tests in TABLE V.

4.5.2.2 Acceptance. Where any EPU fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subjected to the provisions of MIL-STD-105.

4.6 Methods of inspection.

4.6.1 Test Equipment. The test equipment in Table VI shall be used where specified to perform the required tests.

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TABLE VI. Test equipment

Item/Nomenclature	Type		Mfr. & Model or Part No.
	Standard	Peculiar	
1. Test Set ADS (LRU) (M42)		X	11779002
2. Pneumatic Assembly and Adapters		X	
3. Digital Voltmeter (DVM)	X		Fluke 8800 or equivalent
4. Stop Watch	X		Optional
5. Precision Pressure Test Set	X		Texas Instrument Type 156 or equivalent. Range -1000 ft to +10,000 Ft.
6. Water Manometer	X		30 PA 10 (NSN 6685-00-526-8646) or equivalent.
7. Insulation Resistance Tester 50V, zero to 2o M ohms	X		Optional.
8. Weighing machine capable of weighing 7 pounds 9 ounces.	X		Optional.
9. Precision Pressure Gage	X		Texas Instrument Type 145 or equivalent. Range zero to 150 knots.
10. Supply of clean dry air or nitrogen pressure regulated over the range zero to 255 knots.	X		Optional.
11. Milliohmeter	X		Hewlett Packard 4328A or equivalent.

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4.6.2 Performance tests. Using the milliohmeter, verify that the D.C. resistance between the EPU bonding point (any of the EPU feet) and the EPU connector shell is less than 5 milliohms. The EPU should be connected to the Test Set, ADS (LRU) by the appropriate cable harness as shown in Figure 6. Connection of the EPU to the Test Set, ADS (LRU) and Pressure Reference Equipment is as shown in Figure 7. Switch S11 must be in the UP position (Bytes 0-7). Power required by the EPU and the Test Set ADS (LRU) when under test shall be within the steady state limits defined by MIL-STD-704 and shall be less than:

28 Volts	DC	2 Amps
115 Volts	400 Hz	0.5 Amp

Connect the EPU under test to the Test Set, ADS (LRU) as shown in Figure 7. Set S21 on the Test Set to the UP position (ON). Set S22 on the Test Set to ON, and simulteneously operate the STOP WATCH. The ADS FAIL and ADS NO-GO lamps on the Test Set ADS (LRU) shall extinguish within 5 seconds of operating S22. A multimeter set to read dc current shall be connected between AMPS terminals on the POWER TEST panel of the Test Set, ADS (LRU). Set S13 to EPU +28V and depress S14. The multimeter shall indicate a current of 1.4 $\pm$ 0.1 Amps.

4.6.2.1 Operational tests. Throughout the following operational tests, except in paragraph 4.6.2.5.1, ensure that the AADS angle dials B1 and B2 on the Test Set ADS (LRU) are always set such that the difference between their settings is less than 65 degrees. If the difference between settings is more than 65 degrees an AADS Failure, AADS FAIL and ADS NO-GO will be indicated which can only be cancelled by switching S22 on the Test Set, ADS (LRU) off, setting the resolvers B1 and B2 on the test set such that their difference is less than 65 degrees, setting the test set switch S22 to on and then pressing the RESET switch on the front panel of the EPU. Also ensure that switch S5 on the Test Set, ADS (LRU) is kept set to CONT.

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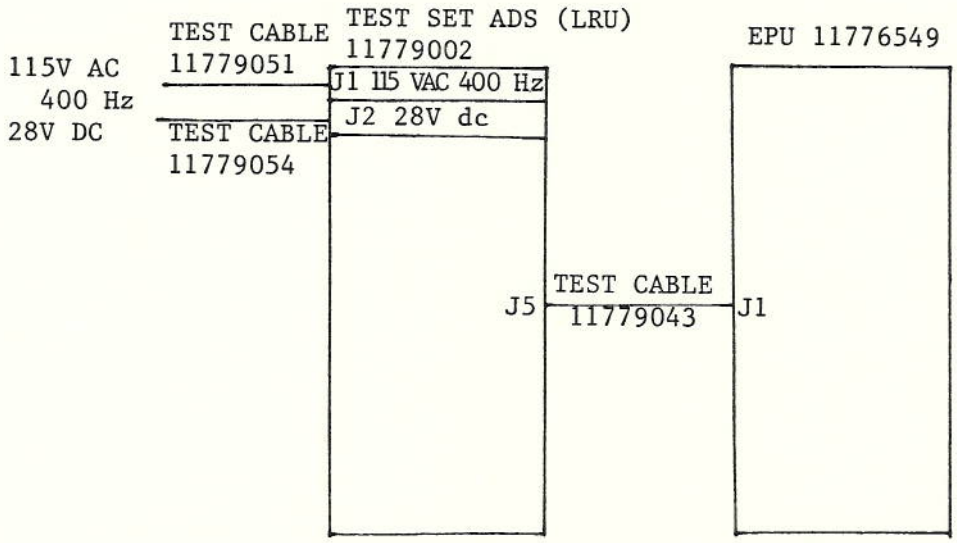


FIGURE 6. Test set, ADS (LRU) - EPU test connection - electrical

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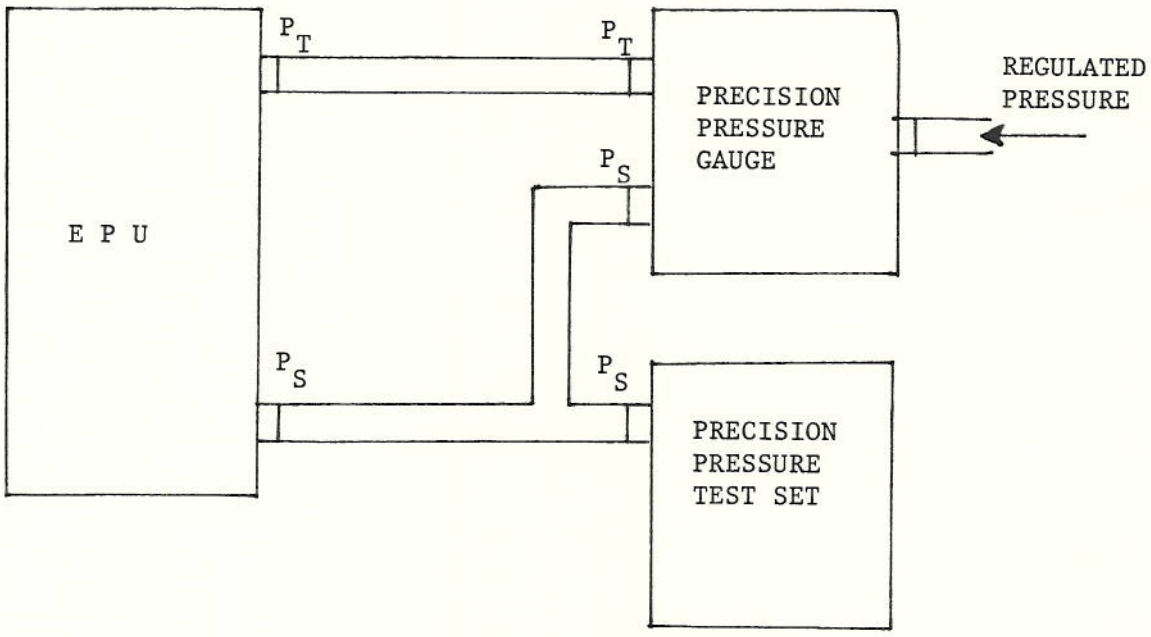


FIGURE 7. Pneumatic connections to pressure test equipment

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4.6.2.2 Pitot pressure leak test. The EPU shall be connected for test as shown in Figures 6 and 8. The AADS angle dials B1 and B2 on the Test Set, ADS (LRU) shall both be set to 0 degrees. The parameter V_u shall be selected on switch S8 for display on the Test Set ADS (LRU). The pressure shall be adjusted to simulate an airspeed of 175 knots. Isolate the EPU from the pressure supply by closing valve A and start the stopwatch. Any leakage displayed on the Test Set, ADS (LRU) shall not exceed 5 units on the Data Word display in three minutes. Reset pitot pressure to zero.

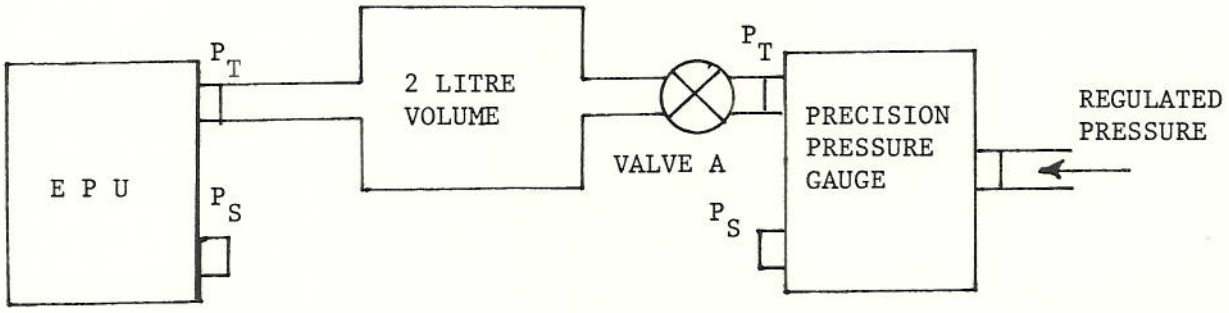


FIGURE 8. Pneumatic circuit - pitot pressure leak test

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4.6.2.3 Static pressure leak test. Connect the EPU for test as shown in Figures 6 and 9. Select the parameter P_{ABS} (S8) for display on the Test Set, ADS (LRU). (Ensure S11 is in the UP position (Bytes 0-7). The pressure test set shall be set to simulate an altitude of 10,000 feet. Isolate the EPU from the static supply by closing valve A and start the stopwatch. Any leakage displayed on the Test Set, ADS (LRU) shall not exceed 2 units on the Data Word Display in 3 1/2 minutes. Do not disconnect the pipes connecting the pressure test set to the EPU. Return Static pressure to ambient. Disconnect the pipes.

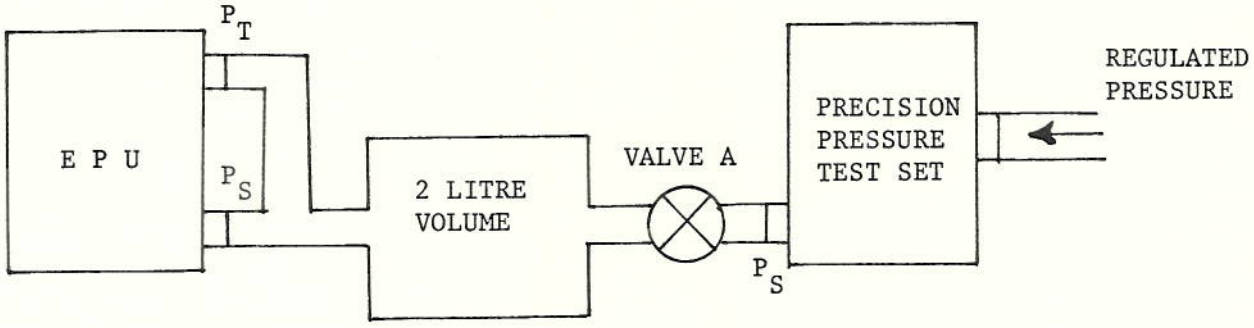


FIGURE 9. Pneumatic circuit - static pressure leak test

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4.6.2.4 Self test. Connect the EPU for test as shown in Figure 6. Set the CPU Test Switch, S1 on the Test Set, ADS (LRU) to the DOWN position. The EPU Test Buttons SELF TEST and RESET shall be operated as specified in Columns 1 and 2 of Table V, and the resultant EPU and Test Set, ADS (LRU) displays shall be as defined in Columns 3 to 8 of Table V.

TABLE V. Self test

Step No.	COL. 1	COL. 2	COL. 3	COL. 4	COL. 5	COL. 6	COL. 7	COL. 8
	Action		Resultant Display					
	EPU Buttons		EPU				Test Set ADS (LRU)	
			Self Test Lamp	Status Indicators			ADS FAIL	ADS NO GO
	SELF TEST	RESET		EPU	AADS	LAI		
1		PRESS & RELEASE	OFF	BLACK	BLACK	BLACK	OFF	OFF
2	PRESS		ON	WHITE	WHITE	WHITE	ON	ON
3	RELEASE		ON	WHITE	WHITE	WHITE	OFF	OFF
4		PRESS & RELEASE	ON	BLACK	BLACK	BLACK	ON	ON
5			OFF	BLACK	BLACK	BLACK	OFF	OFF

The change from Step 4 to Step 5 will occur approximately 15 seconds after initiating Step 4.

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4.6.2.5 CPU test. Connect the EPU for test as shown in Figure 4 and connect the DVM to the terminals below S10 on the Test Set, ADS (LRU). Switch S1 on the Test Set, ADS (LRU) to CPU test and press first the SELF TEST and then RESET buttons on the EPU under test. The ADS FAIL and ADS NO-GO lamps on the Test Set, ADS (LRU) shall illuminate. Switch S10 on the Test Set, ADS (LRU) to the V_{IV} position and verify that the DVM reads -3.33 $\pm$ 0.1V.

4.6.2.5.1 Switch S1 on the Test Set, ADS (LRU) to the DOWN position. The ADS FAIL and ADS NO-GO lamps on the Test Set, ADS (LRU) shall extinguish within two seconds.

4.6.2.6 Monitoring. Select the parameter STS1 on switch S8 of the Test Set, ADS (LRU). The displayed value shall be 000 or 001. Select the parameter V_{DW} on switch S8 of the Test Set, ADS (LRU). The PARITY 1 FAIL lamp shall be extinguished. Select the parameter STS2 on switch S8 of the Test Set, ADS (LRU). The displayed value shall be 000 and the PARITY 2 FAIL lamp shall be extinguished. Press the LAI FAIL switch S2 on the Test Set, ADS (LRU) and hold for at least two seconds before releasing. The LAI Failure Annunciator on the EPU shall change from BLACK to WHITE and stay WHITE. Set the AADS Angle dial B1 to zero degrees and B2 to 60 degrees. The AADS status indicator on the EPU shall be BLACK and the ADS NO-GO and ADS FAIL lamps on the Test Set, ADS (LRU) shall be extinguished. Set the AADS Angle dial B2 to 70 degrees. The AADS status indicator on the EPU shall be WHITE and the ADS NO-GO and ADS FAIL lamps on the Test Set, ADS (LRU) shall be illuminated. Set switch S22 on the Test Set, ADS (LRU) to OF and set AADS angle dial B2 to zero degrees. Set S22 to ON again, wait at least 5 seconds, then press and release the RESET switch on the EPU. The ADS NO-GO and ADS FAIL lamps on the Test Set, ADS (LRU) shall be extinguished and the AADS and LAI status indicators on the EPU shall be BLACK.

4.6.2.7 Air temperature calibration. Connect the EPU for test as shown in Figure 4. Switch S6 on the Test Set, ADS (LRU) as detailed in column 1 of Table VI. Select the parameter T_{AIR} (S8) on the Test Set, ADS (LRU). The Air Temperature Outputs for each test point shall be compared with those values shown in columns 2 and 3 of Table VI.

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TABLE VI. Air temperature calibration

Test Point	Col. 1	Col. 2	Col. 3
	Air Temperature Setting S6	Air Temperature Output (T_{AIR})	
	°C	Nominal	Tol.
1	-40	30	±6
2	+15	140	±6
3	+60	230	±6

4.6.2.8 Radar altimeter. Connect the EPU for test as shown in Figures 6 and 10 and connect the DVM to the terminals below S10. Set the water manometer to 30 Knots. On the Test Set, ADS (LRU) set B1 to 80°, B2 to 60°, and S6 to +15°C, and S10 to V_{IU}. Set S7 to 25 ft. The DVM shall read +0.61V ± 0.4V. Set S7 to 15 ft. The DVM shall read +1.13V ± 0.4V. Depress and hold S3. The DVM shall read +0.61V ± 0.4V. Release S3. Return the water manometer to 0 Knots and set S7 to 30 ft.

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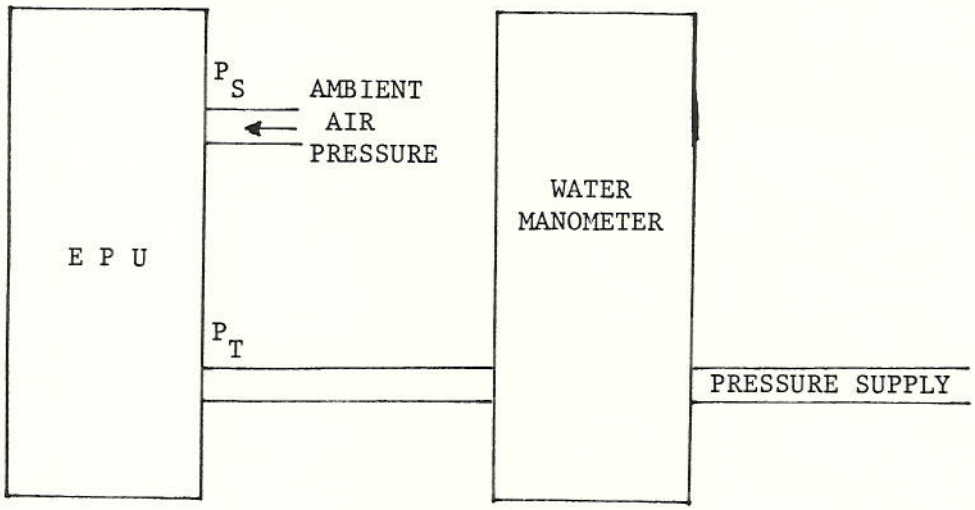


FIGURE 10. Pneumatic connection to water manometer

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4.6.2.9 Indicated airspeed calibration. Connect the EPU for test as shown in Figures 6 and 10. Set the Water Manometer to values defined in column 1 of Table VII. Set the AADS angle dials B1 and B2 on the Test Set, ADS (LRU) as defined in columns 2 and 3 of Table VII. Set switch S7 to +30 ft and S6 to +15°C. The DVM shall be connected to terminals below S10 on the Test Set, ADS (LRU). Select the parameter V_{IU} on Switch S10 of the Test Set, ADS (LRU). The DVM readings for each test point shall be compared with those values shown in columns 4 and 5 of Table VII. Select the Parameter V_{IV} on switch S10 of the Test Set, ADS (LRU). The DVM readings for each test point shall be compared with those values shown in columns 6 and 7 of Table VII. Return Pitot pressure to 0 Knots.

TABLE VII. Indicated airspeed calibration

TEST POINT	COL. 1	COL. 2	COL. 3	COL. 4	COL. 5	COL. 6	COL. 7
	PRESSURE INPUT	AADS Angle Setting ±1°		IAS Outputs			
		α_p	$\alpha_p - \beta_p$	V_{IU} Volts		V_{IV} Volts	
	Knots	Deg.	Deg.	Nominal	Tol.	Nominal	Tol.
1	30	80	60	+0.61	±0.4	+0.70	±0.3
2	35	130	120	-1.81	±0.4	+0.72	±0.3
3	35	70	100	+0.16	±0.4	-1.35	±0.3
4	40	40	50	+1.74	±0.4	-1.11	±0.3
5	60	20	25	+5.00	±0.4	-0.92	±0.3

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4.6.2.10 Altitude Calibration. Connect the EPU for test as shown in Figures 6 and 7. The pressure test set shall be set as detailed in columns 1 and 2 of Table VIII. Set switch S7 to +30 ft.

Note: Care must be taken to maintain the PT-PS pressure between -10mb and +100mb. This pressure is most easily monitored using the precision pressure gauge (Texas Instruments Type 145 or equivalent). In the event of the PT-PS pressure exceeding this range, the ADS NO-GO and ADS FLR lamps on the Test Set, ADS LRU will illuminate, and the EPU status indicator on the EPU under test will show white (FAIL). To cancel these indications, turn test set switch S22 off and on again, then depress RESET switch on the front panel of the EPU under test. Select parameter P_{ABS} (S8) for display on the Test Set, ADS (LRU). The absolute pressure for each test point shall be compared with those values shown in columns 3 and 4 of Table VIII. DO NOT REMOVE the PT-PS pipes. Return static pressure to ambient.

TABLE VIII. Altitude calibration

TEST POINT	COL. 1	COL. 2	COL. 3	COL. 4
	Pressure Input		Absolute Pressure P_{ABS}	
	Ft.	mb	Nominal	Tol.
1	-1000	1050.41	224	±3
2	-0	1013.25	203	±3
3	+4000	875.10	123	±3
4	+8000	752.62	52	±3

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4.6.2.11 True airspeed calibration. Connect the EPU test as shown in Figures 6 and 7. The altitude shall be set as detailed in Column 1 of Table IX. Set the airspeed as detailed in Column 2 of Table IX. Set switch S6 on the Test Set, ADS (LRU) as detailed in column 3 of Table IX. Set the AADS angle dials B1 and B2 as detailed in columns 4 and 5 of Table IX. Set switch S7 to +30 ft. Connect the DVM to S10 terminals on the EPU TEST panel of the Test Set, ADS (LRU). Select the parameters V_u, V_v, and V_w in turn on switch S10 of the Test Set, ADS (LRU). The DVM readings at each test point shall be compared with those values shown in column 6 to 11 of Table IX. Select the parameters V_u, V_v, V_w, and V_{DW} in turn on switch S8 of the Test Set, ADS (LRU). The displayed values shall be compared with those values shown in columns 12 to 19 of Table IX. DO NOT REMOVE the PT-PLS pipes. Return static pressure to ambient and pitot pressure to 0 Knots.

TABLE IX. True airspeed calibration

TEST POINT	COL. 1	COL. 2	COL. 3	COL. 4	COL. 5	COL. 6	COL. 7	COL. 8	COL.9
	Pressure Inputs		Air Temp. Setting	AADS Angle Setting ±1°		Analog TAS Outputs			
				α_p	α_p-β_p	V_u Volts		V_v Volts	
	Ft.	Kts.	°C	Deg.	Deg.	Nom.	Tol.	Nom.	Tol.
1	0	30	15	80	60	+0.30	+0.2	+0.35	±0.15
2	1,000	35	0	130	120	-0.9	+0.2	+0.36	±0.15
3	5,000	100	15	355	355	+5.58	+0.2	-0.14	±0.15
4	10,000	150	0	10	10	+8.61	+0.2	-0.10	±0.15

TEST POINT	COL. 10	COL.11	COL.12	COL.13	COL.14	COL.15	COL.16	COL.17	COL.18	COL.19
	Analog TAS Outputs		Digital TAS Outputs							
	V_w Volts		V_u		V_v		V_w		V_{DW}	
	Nom.	Tol.	Nom.	Tol.	Nom.	Tol.	Nom.	Tol.	Nom.	Tol.
1	0	±0.15	38	±4	156	±12	160	±12	111	±24
2	0	±0.15	14	±4	157	±12	160	±12	105	±24
3	+0.65	±0.15	144	±4	117	±12	212	±12	0	±12
4	-1.37	±0.15	204	±4	120	±12	51	±12	0	±12

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4.6.2.12 LAI power supply test. Connect the DVM to S10 terminals on the EPU TEST panel of the Test Set, ADS (LRU). Select the parameter LAI +15V on switch S10 of the Test Set, ADS (LRU). The DVM reading shall be -15.0 ± 1.0V dc. Select the parameter LAI ±15V on switch S10 of the Test Set, ADS (LRU). The DVM reading shall be -15.0 ± 1.0V dc. Turn OFF the Test Set. Disconnect all equipment.

4.6.3 Physical characteristics.

4.6.3.1 Dimensions. Use standard practices to ensure the requirements of drawing are met.

4.6.3.1.2 Weight. Use item 8 of Table VI to ensure that requirements of 3.4.3.2 are met.

4.6.3.2 Preconditioning (burn-in). Subject each EPU to the burn-in requirements of Appendix B.

4.6.4 Environmental tests.

4.6.4.1 Natural environment. Subject the EPU to the following environments.

4.6.4.2 Altitude. The EPU shall be subjected to the tests of Method 504.1 of MIL-STD-810 for Category 4 equipment. The EPU, nonoperating, shall be subjected to Method 500.1, Procedure I, with pressures simulating air transport at 50,000 feet altitude at a pressure of 2.85 inches of mercury (Hg) for 1 hour. At the conclusion of this test, subject the EPU to the performance tests of Table IV.

4.6.4.3 Temperature. The EPU shall be subjected to test Method 501.1, Procedure II, of MIL-STD-810, except Step 4 temperature shall be 71°C. The system shall then be subjected to the test of Method 502.1, Procedure I, except maximum low temperature shall be -54°C. At the conclusion of this test, subject the EPU to the performance tests of Table IV.

4.6.4.4 Humidity. The EPU shall be subjected to test Method 507.1, Procedure II, of MIL-STD-810. In Step 6, measurements shall be taken during the last five hours of the last test cycle. At the conclusion of this test, subject the EPU to the performance tests of Table IV.

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4.6.4.5 Sand and dust. The EPU shall be subjected to test Method 510.1, Procedure I, of MIL-STD-810, except that the air velocity in Steps 1, 2, and 3 shall be 200 ± 100 FPM. Step 3 may proceed immediately after stabilization in Step 2, if desired. At the conclusion of this test, subject the EPU to the performance tests of Table IV.

4.6.4.6 Salt atmosphere. The EPU shall be subjected to the test Method 509.1 of MIL-STD-810. Operation of the EPU shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surface.

4.6.4.7 Fungus. The EPU shall be subjected to the test of Method 508.1, Procedure I, of MIL-STD-810. Gasket sealed or other non-hermetic sealed test items shall be opened during test exposure and all internal and external surfaces shall be sprayed with spore suspension. Operation of the equipment is not required, but examination of the concentration after exposure period shall reveal no evidence of viable fungus on any surfaces.

4.6.4.8 Explosive atmosphere. The EPU shall be subjected to test Method 511.1, Procedure I, of MIL-STD-810, except test shall be limited to a maximum altitude of 20,000 feet. At the conclusion of this test, subject the system to the performance tests of Table IV.

4.6.4.9 Rain. The EPU shall be subjected to test Method 506.1, Procedure I, of MIL-STD-810. During each thirty minute rain period, the wind shall be maintained for 15 minutes, commencing 5 minutes after the start of the period as specified in the procedures. The system shall be operated during the last 10 minutes of the thirty minute period. At the conclusion of this test, subject the EPU to the performance tests of Table IV.

4.6.4.10 Service environment.

4.6.4.10.1 Bench handling. The system shall be subjected to Method 516.2, Procedure V, of MIL-STD-801. At the conclusion of this test, subject the EPU to the performance tests of Table IV.

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5. PACKAGING.

5.1 Preservation, packing, packaging, and marking. The EPU shall be preserved, packaged, packed, and marked in accordance with the requirements of the preservation and packaging data sheet for P11776549. At the level of protection specified in the contract (see 6.2b).

5.2 Commercial packaging shall be in accordance with ASTM-D-3951.

6. NOTES.

6.1 Intended use. This equipment is intended for use in the AH-1S (modernized) helicopters.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging Data Sheet (see 5.1).

c. Selection of applicable levels of preservation, packaging, and packing.

d. Applicable stock number.

e. First article inspection requirements.

6.3 Previous revisions of MIL-P-70486 (AR) were for the US Army Armament Research and Development Center internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1270-A054)

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APPENDIX A

Signal Interface Requirements — Ident Number 1

Power Signal Interface Requirements

Nomenclature	AADS REF
Type and Frequency	AC 400 ± 20 Hz
Conductor Type and Gauge	Twisted Shielded pair 24 AWG
Origin	EPU
Destination	AADS
Destination Impedence	> 300 ohms
Source	EPU
Source Impedence	< 300 ohms
Power Required	180 mW
Phase	N/A
Maximum Current	25 mA
Nominal Voltage	18.45V peak to peak
Maximum Voltage	20.5V peak to peak
Minimum Voltage	12.0V peak to peak
Power Factor	N/A
Regulation	N/A

REMARKS

Excitiation Signal for AADS Angle Resolvers.

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APPENDIX A

Signal Interface Requirements — Ident Number 2

Analog Signal Interface Requirements

Nomenclature	SIN Alpha
Type and Frequency	AC Analog 400 ± 20 Hz
Conductor Type and Gauge	Twisted Shielded Triple [1]24 AWG
Origin	AADS
Destination	EPU
Destination Impedence	> 10K ohms
Source	AADS
Source Impedence	< 300 ohms
Voltage Range	$0.454V_R$ Volts to peak[2]
Scale Factor	$0.454V_R$ SIN Alpha
Parameter Range	+1 to -1
Parameter Accuracy	Equivalent to an angle accuracy of ±2 degrees
Phasing	N/A
RMS Null	0 VDC when Sin Alpha = 0
Phase Shift	17 ± 3 Degrees Lead. Relative to AADS Ref.
Sensitivity	N/A
Maximum Current	N/A
Maximum Voltage	9.6V peak to peak

REMARKS

[1]Sin Alpha, Cos Alpha and Alpha RTN.

[2]V_R = AADS Ref Peak to Peak Voltage.

Angular position signal from AADS Resolver.

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APPENDIX A

Signal Interface Requirements — Ident Number 3

Analog Signal Interface Requirements

Nomenclature	Cos Alpha
Type and Frequency	AC Analog 400 ± 20 Hz
Conductor Type and Gauge	Twisted Shielded Triple [1]24 AWG
Origin	AADS
Destination	EPU
Destination Impedence	>10K ohms
Source	AADS
Source Impedence	< 300 ohms
Voltage Range	$0.454V_R$ Volts to peak[2]
Scale Factor	$0.454V_R$ COS Alpha
Parameter Range	+1 to -1
Phasing	Equivalent to an angle accuracy of ± 2 degrees
RMS Null	0 Vdc when Cos Alpha = 0
Phase Shift	17 ± 3 degrees Lead. Relative to AADS Ref.
Sensitivity	N/A
Maximum Current	N/A
Maximum Voltage	9.6V Peak to Peak

REMARKS

[1]Sin Alpha, Cos Alpha and Alpha RTN.

[2]V_R = AADS Ref Peak to Peak Voltage.

Angular position signal from AADS Resolver.

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APPENDIX A

Signal Interface Requirements — Ident Number 4

Analog Signal Interface Requirements

Nomenclature	Sin (Alpha-Beta)
Type	AC Analog 400 $\pm$ 20 Hz
Conductor Type and Gauge	Twisted Shielded Triple [1] 24 AWG
Origin	AADS
Destination	EPU
Destination Impedence	$>$ 10K ohms
Source	AADS
Source Impedence	$<$ 300 ohms
Voltage Range	$0.454V_R$ Volts to peak [2]
Scale Factor	$0.454V_R$ Sin (Alpha-Beta)
Parameter Range	+1 to -1
Parameter Accuracy	Equivalent to an angle accuracy of $\pm$ 2 degrees
Phasing	N/A
RMS Null	0 Vdc when Sin (Alpha-Beta) = 0
Phase Shift	17 $\pm$ 3 degrees Lead. Relative to AADS Ref.
Sensitivity	N/A
Maximum Current	N/A
Maximum Voltage	9.6V Peak to Peak

REMARKS

[1] Sin (Alpha-Beta), Cos (Alpha-Beta) and Alpha-Beta RTN.

[2] V_R = AADS Ref Peak to Peak Voltage.

Angular position signal from AADS Resolver.

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APPENDIX A

Signal Interface Requirements — Ident Number 5

Analog Signal Interface Requirements

Nomenclature	Cos (Alpha-Beta)
Type	AC Analog 400 $\pm$ 20 Hz
Conductor Type and Gauge	Twisted Shielded Triple [1] 24 AWG
Origin	AADS
Destination	EPU
Destination Impedence	$>$ 10K ohms
Source	AADS
Source Impedence	$<$ 300 ohms
Voltage Range	$0.454V_R$ Volts to peak [2]
Scale Factor	$0.454V_R$ Cos (Alpha-Beta)
Parameter Range	+1 to -1
Parameter Accuracy	Equivalent to an angle accuracy of $\pm$ 2 degrees
Phasing	N/A
RMS Null	0 Vdc when Cos (Alpha-Beta) = 0
Phase Shift	17 $\pm$ 3 degrees Lead. Relative to AADS Ref.
Sensitivity	N/A
Maximum Current	N/A
Maximum Voltage	9.6V Peak to Peak

REMARKS

[1] Sin (Alpha-Beta), Cos (Alpha-Beta) and Alpha-Beta RTN.

[2] V_R = AADS Ref Peak to Peak Voltage.

Angular position signal from AADS Resolver.

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APPENDIX A

Signal Interface Requirements Ident Number 6

Analog Signal Interface Requirements

Nomenclature	Air Temperature (ATS)
Type	Resistance
Conductor Type and Gauge	Twisted Shielded Pair 24 AWG
Origin	AADS
Destination	EPU
Destination Impedence	N/A
Source	AADS
Source Impedence	477.17 ohms at $0^{o}C$
Voltage Range	N/A
Scale Factor	Approx 2 ohms/^{o}C
	$Rt = 477.17\ (1+.0039\ 58t-.000000583t^{2})$
	where Rt resistance in ohms at $t^{o}C$
Parameter Range	374.4 to 609.9 ohms
Parameter Accuracy	0.6 ohms at $0^{o}C$
Phasing	N/A
RMS Null	N/A
Phase Shift	N/A
Sensitivity	N/A
Maximum Current	5mA
Maximum Voltage	3.05V

REMARKS

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APPENDIX A

Signal Interface Requirements — Ident Number 7

Analog Signal Interface Requirements

Nomenclature	Fore-Aft Indicated Airspeed (VIU)
Type	DC Analog
Conductor Type and Gauge	Twisted Shielded Pair
Origin	EPU
Destination	LAI
Destination Impedence	>10K ohms
Source	EPU
Source Impedence	<300 ohms
Voltage Range	-5.0Vdc to +5.0Vdc
Scale Factor	100 Millivolts per Knot
Parameter Range	-50 to +50 Knots[1]
Parameter Accuracy	± 4 Knots (2 σ)
Phasing	N/A
RMS Null	N/A
Phase Shift	N/A
Sensitivity	N/A
Maximum Current	N/A
Maximum Voltage	-5.0Vdc +5.0Vdc

REMARKS

[1]Positive voltage corresponds to Headwind/Forward aircraft flight.

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APPENDIX A

Signal Interface Requirements — Ident Number 8

Analog Signal Interface Requirements

Nomenclature	Lateral Indicated Airspeed (VIV)
Type	DC Analog
Conductor Type and Gauge	Twisted Shielded Pair
Origin	EPU
Destination	LAI
Destination Impedence	>10K ohms
Source	EPU
Source Impedence	<300 ohms
Voltage Range	-5.0Vdc to +5.0Vdc
Scale Factor	100 Millivolts per Knot
Parameter Range	-50 to +50 Knots
Parameter Accuracy	± 3 Knots (2 σ)
Phasing	N/A
RMS Null	N/A
Phase Shift	N/A
Sensitivity	N/A
Maximum Current	N/A
Maximum Voltage	± 5Vdc

REMARKS

[1]Positive voltage corresponds to Rightward Flight or Sidewind from the right.

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APPENDIX A

Signal Interface Requirements Ident Number 9

Discrete Signal Interface Requirements

Nomenclature	LAI Failure (ADS FLR/BAR)
Type	Discrete
Conductor Type and Gauge	Single Unshielded
Origin	EPU
Destination	LAI
Destination Impedence	> 1K ohms
Source	EPU
Source Impedence	True State > 10K ohms
	False State < 25 ohms
Maximum Current	5 mA
Maximum Voltage	+15Vdc
True State Voltage	Open Circuit[1]
False State Voltage	0V ± 0.2V

REMARKS

[1]Output is pulled up to +15V by LAI

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APPENDIX A

Signal Interface Requirements — Ident Number 10

Discrete Signal Interface Requirements

Nomenclature	LAI Failure (LAI FLR/BAR)
Type	Discrete
Conductor Type and Gauge	Single Unshielded
Origin	EPU
Destination	LAI
Destination Impedence	>1K ohms
Source	EPU
Source Impedence	True State: > 10K ohms
	False State: < 25 ohms
Maximum Current	5 mA
Maximum Voltage	+5Vdc
True State Voltage	Open Circuit[1]
False State Voltage	0V ± 0.2V

REMARKS

[1]Input is pulled up to +5V by EPU

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APPENDIX A

Signal Interface Requirements — Ident Number 11

Power Signal Interface Requirements

Nomenclature	±15V LAI
Type	DC
Conductor Type and Gauge	Twisted Pair 22 AWG
Origin	EPU
Destination	LAI
Destination Impedence	N/A
Source	EPU
Source Impedence	N/A
Power Required	N/A
Phase	N/A
Maximum Current (Total)	267 mA
Nominal Voltage	±15Vdc
Maximum Voltage	±16Vdc
Minimum Voltage	±14Vdc
Power Factor	N/A
Regulation	N/A

REMARKS

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APPENDIX A

Signal Interface Requirements — Ident Number 73

Discrete Signal Interface Requirements

Nomenclature	ADS Go/No Go
Type	Discrete
Conductor Type and Gauge	Single Unshielded
Origin	
Destination	Fire Control Computer & Caution Warning
Destination Impedance	10K OHMS ±20%
Source	Air Data Subsystem
Source Impedance	< 10 OHMS
Max Current	N/A
Max Voltage	MIL-STD-704
True State Voltage	+28 VDC
False State Voltage	Open (No Go)

REMARKS

2 ma. For activating caution subsystem and FCC.

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APPENDIX A

Signal Interface Requirements — Ident Number 74

Digital Signal Interface Requirements

Nomenclature	Data Clock (with complement)
Type	Square Wave
Conductor Type and Gauge	Shielded Twisted Pair
Origin	Aircraft
Destination	Air Data Subsystem
Destination Impedance	7820 or Equivalent
Source	Fire Control Computer
Source Impedance	7830 or Equivalent
Address	N/A
No. of Bits/Words	N/A
First Transmitted Bit	N/A
Scale Factor	N/A
LSB Value (Resolution)	N/A
Parameter Range	N/A
Parameter Accuracy	N/A
Transmission Rate	10 KHz Rate (Nominal)
Iteration Rate	N/A
True State Voltage	TTL Logic Level 1[1]
False State Voltage	TTL Logic Level 0[1]
Min Sample Rate	N/A

REMARKS

[1] Signal complement has logic -0 true state and logic -1 false state.

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APPENDIX A

Signal Interface Requirements — Ident Number 75

Digital Signal Interface Requirements

Nomenclature	ADS Data (with complement)
Type	Serial Digital
Conductor Type and Gauge	Shielded Twisted Pair
Origin	Air Data Subsystem
Destination	Fire Control Computer
Destination Impedance	7820 or Equivalent
Source	Electronics Processor Unit
Source Impedance	7830 or Equivalent
Address	N/A
No. of Bits/Words	(6) 8-Bit Word-Periods (Parity Bit 49)
First Transmitted Bit	LSB or Word 1
Scale Factor	As in Table I
LSB Value (Resolution)	As in Table I
Parameter Range	As in Table I
Parameter Accuracy	As in Table I
Transmission Rate	10 KHz Nominal
Iteration Rate	5 HZ
True State Voltage	TTL Logic 1[1]
False State Voltage	TTL Logic 0[1]
Voltage Range	TTL Levels[1]

REMARKS

[1] Signal complement has logic -0 true state and logic -1 false state.

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APPENDIX A

Signal Interface Requirements Ident Number 77

Digital Signal Interface Requirements

Nomenclature	Data Ready (and Return)
Type	Digital
Conductor Type and Gauge	Shiedled Pair
Origin	
Destination	Fire Control Computer
Destination Impedance	7820 Receiver or Equivalent
Source	Air Data Subsystem
Source Impedance	7830 Driver or Equivalent
Address	N/A
No. of Bits/Words	N/A
First Transmitted Bit	N/A
Scale Factor	N/A
LSB Value (Resolution)	N/A
Parameter Range	N/A
Parameter Accuracy	N/A
Transmission Rate	N/A
Iteration Rate	N/A
True State Voltage	TTL Logic Level 1
False State Voltage	TTL Logic Level 0

REMARKS

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APPENDIX A

Signal Interface Requirements — Ident Number 101

Analog Signal Interface Requirements

Nomenclature	Fore-Aft True Airspeed
Type	DC Analog
Conductor Type and Gauge	Twisted Shiedled Pair
Origin	
Destination	Doppler Navigation Set
Destination Impedance	$\leq$ 10 K Ohms
Source	Air Data Subsystem
Source Impedance	$<$ 10 Ohms
Voltage Range	-1.75 Vdc to +10 Vdc
Scale Factor	50 Millivolts Per Knot
Parameter Range	-35 to +200 Knots[1]
Parameter Accuracy	$\pm$4 Knots (2 σ)
Phasing	N/A
RMS Null	N/A
Phase Shift	N/A
Sensitivity	N/A
Max Current	10 ma
Max Voltage	-2.5, +10.0 Vdc

REMARKS

[1] Positive Voltage Corresponds to Headwind/Forward Aircraft Flight.

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APPENDIX A

Signal Interface Requirements — Ident Number 102

Analog Signal Interface Requirements

Nomenclature	Lateral True Airspeed
Type	DC Analog
Conductor Type and Gauge	Twisted Shiedled Pair
Origin	
Destination	Doppler Navigation Subsystem
Destination Impedance	$\geq$ 10 K Ohms
Source	Air Data Subsystem
Source Impedance	$<$ 10 Ohms
Voltage Range	$\pm$2.5 Vdc
Scale Factor	50 Millivolts Per Knot
Parameter Range	-50 to +50 Knots[1]
Parameter Accuracy	$\pm$3 Knots (2 σ) Out to $\pm$ 35 Knots Limit
Phasing	N/A
RMS Null	N/A
Phase Shift	N/A
Sensitivity	N/A
Max Current	10 ma
Max Voltage	$\pm$2.5 Vdc

REMARKS

[1] Positive Voltage Corresponds to Rightward Flight or Sidewind from the Right.

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APPENDIX A

Signal Interface Requirements — Ident Number 103

Analog Signal Interface Requirements

Nomenclature	Up-Down True Airspeed
Type	DC Analog
Conductor Type and Gauge	Twisted Shiedled Pair
Origin	
Destination	Doppler Navigation Subsystem
Destination Impedance	$\geq$ 10 K Ohms
Source	Air Data Subsystem
Source Impedance	$<$ 10 Ohms
Voltage Range	-2 Vdc to +1 Vdc
Scale Factor	50 Millivolts Per Knot
Parameter Range	-40 to +20 Knots[1]
Parameter Accuracy	$\pm$3 Knots (2 σ)
Phasing	N/A
RMS Null	N/A
Phase Shift	N/A
Sensitivity	N/A
Max Current	10 ma
Max Voltage	-2, +1 Vdc

REMARKS

[1] Positive Voltage Corresponds to Upward Air Velocity or Descending Flight.

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APPENDIX A

Signal Interface Requirements — Ident Number 114

Digital Signal Interface Requirements

Nomenclature	Reliability
Type	DC
Conductor Type and Gauge	Twisted Shiedled Pair
Origin	
Destination	Air Data Subsystem
Destination Impedance	$\geq$ 2 K Ohms
Source	Radar Altimeter
Source Impedance	$\leq$ 200 Ohms
Address	N/A
No. of Bits/Words	N/A
First Transmitted Bit	N/A
Scale Factor	N/A
LSB Value (Resolution)	N/A
Parameter Range	N/A
Parameter Accuracy	N/A
Transmission Rate	N/A
Iteration Rate	N/A
True State Voltage	4.5 $\pm$1.0 Vdc During Track
False State Voltage	0.5 $\pm$0.5 Vdc During Search or No Track

REMARKS

The Signal Confirms Validity of Radar Altimeter Signal.

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APPENDIX A

Signal Interface Requirements Ident Number 132

Power Signal Interface Requirements

Nomenclature	28 Vdc Operating Power and Return
Type	DC
Conductor Type and Gauge	22 AWG
Origin	Aircraft
Destination	Air Data Subsystem
Destination Impedance	N/A
Source	Aircraft Power
Source Impedance	N/A
Power Required	30 Watts
Phase	N/A
Maximum Current	1.2 Amp
Nominal Voltage	28V
Maximum Voltage	Per MIL-STD-704
Minimum Voltage	Per MIL-STD-704
Power Factor	N/A
Regulation	N/A
Voltage Range	Per MIL-STD-704

REMARKS

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APPENDIX A

Signal Interface Requirements Ident Number 133

Power Signal Interface Requirements

Nomenclature	LAI Lighting Power and Return
Type	DC
Conductor Type and Gauge	22 AWG
Origin	
Destination	LAI
Destination Impedance	N/A
Source	Pilot's Station Controls
Source Impedance	
Power Required	5 Watts
Phase	N/A
Maximum Current	1 Amp
Nominal Voltage	5 Volts, Dimmable
Maximum Voltage	5 Vdc
Minimum Voltage	0 Vdc
Power Factor	N/A
Regulation	N/A

REMARKS

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APPENDIX B

BURN-IN REQUIREMENTS

Burn-in is to be applied to all deliverable systems or LRU's.

Test Environments

Environmental exposure for the EPU during burn-in is Level F of MIL-STD-781B. Cycle time shall be determined by thermal survey. Chamber temperature change shall be programmed equal to or greater than 5° per minute. Figure 2 illustrates one cycle of the environmental profile described below.

Temperature	-50°C to +71°C (-65°F to 160°F).
Temperature Cycling	Temperature cycling shall be time to stabilize at low temperature followed by time to stabilize at the high temperature, plus 2 hours.
Vibration	2.2G ± 10% peak acceleration value at 30 ± 5 Hz measured at the mounting points. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time.
Equipment On-Off Cycling	Equipment off during cooling cycle and on during heating cycle.
Input Voltage	Specified voltage +5 -2%.
Input Voltage Cycling	a. Nominal voltage for first 1/2 of "ON" time starting at the beginning of the heating period. b. Momentary off. c. 100% of nominal for next 1/3 of "ON" time. d. Monemtary off. e. 90% of nominal for last 1/3 of "ON" time.

NOTE: Voltage Cycling will occur on the primary EPU power (28 VDC) only.

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APPENDIX B

Number of Cycles

The minimum number of cycles shall be five (5) with the last two (2) cycles failure free.

Duty Cycle Information

During equipment ON time, mode selection time shall be apportioned as follows:

Test	95% ± 5%

During operation, the display and BIT Indicators shall be monitored.

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